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Editorial

Agriculture, structural transformation and poverty reduction: Eight new insights

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ABSTRACT

Whether the sector of growth matters for the speed of poverty reduction, why, and how best to invest to maximize the poverty reducing effects of sectoral growth remain topics of intense debate. Drawing on the more recent history and applying a range of methods, the papers in this special issue confirm the view that growth in agriculture is on average more poverty reducing than an equivalent amount of growth outside agriculture. They also add important nuances to this broad empirical regularity, uncover a series of structural conditions that affect this relation, and show that different mechanisms to finance public investment to boost sectoral growth (deficit financing, taxation or aid) can have widely different impacts on poverty, a widely-ignored issue so far. They do so by going beyond the traditional agriculture-nonagriculture dichotomy, also looking at the subsectors and the differential effects of prices versus productivity. They further distinguish between production for home and market consumption, between modern, outward oriented and informal domestically - focused firms, as well as between secondary towns and cities. Eight key insights emerge.

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1. Introduction

Whether the sector of growth matters for the speed of poverty reduction, and why, has been the topic of intense study and debate. The debate was especially lively during the 1990s and 2000s, in the run-up to the 2008 world food price crisis. With the adoption of the Millennium Development Goals, the world had shifted its focus to poverty reduction, world food prices had been on a three-decade downward trend, and agriculture was no longer considered so critical to poverty-reduction efforts as during the 1960s and 1970s (Johnston & Mellor, 1961; Timmer, 2010). At the same time, and drawing on recent history, a growing number of empirical studies confirmed nonetheless that growth in agriculture had been more effective in reducing poverty over the past decades than growth outside agriculture.[1]

Yet, the continuing relevance of these findings to support a dominant role for agricultural growth in poverty reduction going forward was also questioned. Growth had been strong in many developing countries, and the share of agriculture in these economies had declined. Economies had become more open, following globalization, making food more tradable and reducing the relevance of insights from closed economy models. And the dismal performance of African agriculture during much of the 1990s and 2000s, did not inspire much hope for reliance on that sector to drive a growth and poverty reduction agenda. In this view, productivity in agriculture, and especially in African smallholder agriculture, is so low compared with the other sectors and food sufficiently tradeable, that poverty reduction is much more likely to come from urbanization, i.e. facilitation of migration out of agriculture, and industrialization, greater reliance on food trade, and a radical transformation of the agricultural sector (into highly mechanized large-scale farming) (Collier and Dercon, 2014; Dercon and Gollin, 2014).

Against this background, the participants in a workshop on "Agriculture, structural transformation and poverty reduction", organized at the World Bank in late 2013, set out to update and fine-tune the empirical evidence on the relationship between sectoral growth and poverty reduction. They further zoomed in on the structural factors and policy environments conditioning the conversion of growth in the different sectors into poverty reduction, with special attention to the role of the structural (and spatial) transformation, i.e. the labor reallocation from (rural) agriculture to (urban) nonagriculture.

Much of the more recent literature on the topic has been econometric, mostly focused on comparing the differences in aggregate poverty effects from growth across the sectors. Given limited sample sizes, it often paid only limited attention to potential heterogeneity in the effects, across agricultural and nonagricultural *sub*sectors, across countries at different levels of development, or even across different poverty outcomes, such as nutrition. Given the econometric focus, it also rarely distinguished between growth resulting from increases in productivity and growth from other sources, such as price changes, and relatedly the possibly substantially different impacts of productivity growth in open and closed

[1] Ravallion and Datt (1996), Datt and Ravallion (1998), Bravo-Ortega and Lederman (2005), Tiffin and Irz (2006), Ravallion and Chen (2007), World Bank (2008), Suryahadi, Suryadarma and Sumarto (2009), Ferreira, Leite and Ravallion (2010), Loayza and Raddatz (2010), Montalvo and Ravallion (2010), Christiaensen, Demery and Kuhl (2011).

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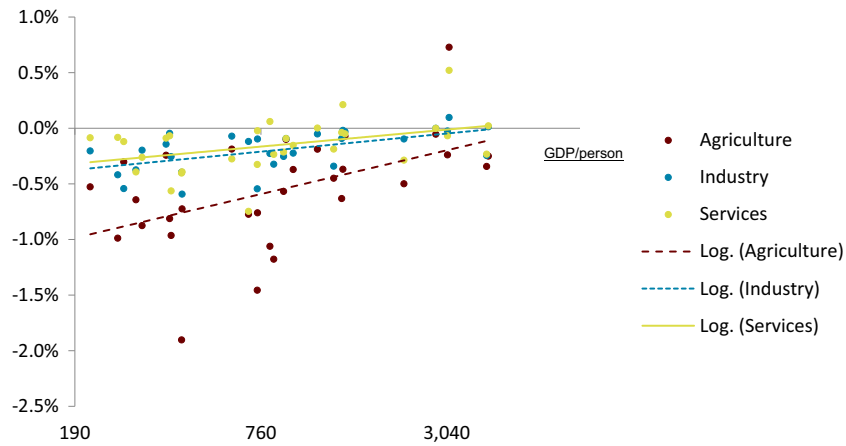

Fig. 1. Relationship between per capita GDP and poverty change from a productivity increase equal to 1% of GDP (single country simulations). *Source:* Ivanic and Martin, 2018.

economies, an important insight highlighted by Matsuyama (1992). The literature further abstracted often from the role of the sectoral interactions (between agriculture and nonagriculture) and their drivers, the effectiveness of different interventions in getting agricultural or nonagricultural growth going, including through structural transformation, and last, but not least, the eventual impact of different financing modalities for poverty.

The eight papers in this special issue, retained after a standard refereeing process coordinated by us under the auspices of the World Development Editor in Chief, begin to address these gaps. Three papers are global/cross-country in scope (Ligon and Sadoulet; Ivanic and Martin; Eberhardt and Vollrath), while the others take a more region-specific lens; four of them focused on Africa (Dorosh and Thurlow; Kirk, Kilic and Carletto; Diao and McMillan; Adam, Bevan and Gollin), and one on a recent successful poverty reducer from South Asia, Bangladesh (Emran and Shilpi). The papers are essentially empirical and policy oriented in spirit. They pay great attention to the magnitudes of the different effects and their sensitivity to different assumptions and are strongly motivated by theory. They are also pluralistic in their analytical approaches, using both econometric estimation techniques and calibrated computable general equilibrium (CGE) model simulations linked with household models to explore the issues. The following are eight broad insights that emerged.

2. Eight new insights

First, the proposition that growth in agriculture is in general (two to three times) more effective at reducing poverty than an equivalent amount of growth generated outside agriculture remains confirmed. This holds irrespective of the empirical method used (econometric, controlling for the endogeneity of growth, as in Ligon and Sadoulet, or CGE model simulations as in Ivanic and Martin or Dorosh and Thurlow). The finding also does not change when considering the distribution of welfare measures *within* countries or *across* countries (Ligon and Sadoulet)[2] or when using poverty to growth semi-elasticities instead of elasticities (Dorosh and Thurlow).[3] This result holds more strongly when the experiment considered is, as in Ravallion and Datt (1996), an increase in GDP of equal size across sectors than an equal increase in GDP growth in each sector, but in poorer economies—where agriculture is a larger share of GDP—it holds for both experiments.

Second, the advantage of agriculture over nonagriculture in reducing poverty is largest for the poorest in society (Ligon and Sadoulet) and ultimately disappears as countries become richer. This is again borne out both by the econometric (Ligon and Sadoulet) and CGE model simulation results (Ivanic and Martin) (Fig. 1). Similarly, Ligon and Sadoulet find stronger progressivity in the poverty reducing effect of agricultural growth over nonagricultural growth when literacy rates are lower, but could not discern any systematic effect of initial inequality.

These two sets of findings are overall consistent with the earlier cross-country findings reported in Christiaensen, Demery and Kuhl (2011) and underscore the continuing role agriculture could play in accelerating poverty reduction, especially in Africa, where the world's poverty is increasingly concentrating, and South Asia, which also still houses one third of the world's extreme poor (Castaneda et al., 2018).

Third, there is substantial heterogeneity in the poverty-reducing effects of nonagriculture across its different subsectors (Dorosh and Thurlow), with poverty to growth elasticities for trade and transport services closer to those of agriculture and those for manufacturing, especially agro-processing, at times even exceeding them. The poverty-reducing effects of an equal amount of growth in mining, on the other hand, but also in finance, and business and government services are much more limited. These results are based on CGE model simulations, thereby accounting for the differential strength of the intersectoral linkages. Increasing the productivity of trade and transport reduces transaction costs for all marketed products for example, which is particularly beneficial for the agents in the sectors whose products have high margins, such as agriculture and food. Second, trade services are often also provided by low-paid informal traders, such that productivity growth in trade and transport services can have both direct and indirect linkages to the poor. Adam, Bevan and Gollin (also in this special issue) find similarly a substantial increase in the incomes of unskilled and rural workers in Tanzania from reducing transaction

[2] One choice to make in cross-country analysis is to consider the effect on the distribution of welfare measures within countries, or across countries. Under the former, it is the effect of sectoral growth on the welfare measures across the deciles within each country that is examined. Under the latter, a global welfare distribution is constructed (using purchasing power parity price corrections) and a global poverty line is used, such that people are considered poor according to globally uniform benchmark. One then looks at the differential effect of sectoral growth on the poor in each country, as defined by this global poverty line.

[3] The semi-elasticity of poverty to growth is the absolute change in the poverty measure per percent change in income. It was introduced by Misselhorn and Klasen (2006) to correct for the large (numeric) sensitivity of the poverty to growth elasticity to the initial level of poverty.

costs in transport, especially when the gains are obtained from a reduction in rents.

Fourth, the advantages of growth in agriculture over growth in nonagriculture in reducing poverty can also extend to other welfare outcomes such as food insecurity and malnutrition, but tend to be more subject to the market context (depth of local food markets), the type of agricultural income growth (for commercial purposes or own consumption) and, when grown for own consumption, depending on the nutritional value of the crop. Recent cross-country evidence by Headey (2013) suggests for example that agricultural growth is also the most effective way to fight malnutrition, possibly because it is more effective at increasing the incomes of the poor, as shown above. Kirk, Kilic and Carletto in this issue further examine this proposition and highlight that not all agriculture will be equally good for reducing poverty or malnutrition. In their analysis from Uganda, focused on nutrition, children's long run nutritional status correlates negatively with the share of income from crop production, especially among the older and poorer subset of children. This follows from the large share of low-protein staple production for own consumption among the households in this subsample (such as cassava and plantain). While they find the effect to be small, the findings underscore the importance of context and the need for further disaggregation within agriculture and non-agriculture, especially when deciding to promote certain sectors to improve nutritional outcomes.

Fifth, the degree of tradability of food (and nonfood) and the range of economies experiencing the increase in productivity are important considerations in determining the advantage of agriculture over non-agriculture in reducing poverty, though CGE simulations for 315,000 households from 31 countries indicate that agriculture's advantage holds empirically, both under open (food is tradable) and closed (food is non-tradable) economy assumptions and when productivity growth is confined to one country or widespread (across all developing or all countries) (Ivanic and Martin). However, the source of the poverty reducing benefits from agricultural productivity growth changes as innovations are more widely adopted, moving from increases in producer returns (and wage labor opportunities) to reductions in consumer prices. The robustness of the gain across degrees of openness raises questions about the conclusion in Dercon (2009) that agricultural growth is a priority for poverty reduction only in landlocked African economies.

Sixth, rising agricultural productivity not only reduces poverty by releasing (agricultural) labor to nonagricultural activities, it can also do so by pulling surplus labor from less productive home production into agriculture (Emran and Shilpi). When agricultural productivity increases, poor households may gain directly as producers, when costs fall more than prices or indirectly, as consumers through lower prices, or as agricultural laborers through increased employment and higher wages. The latter channel is especially important in densely populated countries with well-developed agricultural wage labor markets as in Asia.[4] The poverty reducing potential via the labor market channels depends on the nature of the technology (labor (or land) saving), inducing either an inward (or outward) shift of the agricultural labor demand curve, and thus a decline (increase) in agricultural wages and employment. Emran and Shilpi, further show that the agricultural wage and employment effects do not have to be symmetrical however, and that little or no increase in agricultural wages and a decline in agricultural wage employment may be consistent with poverty reduction, since a substantial increase in household labor supply to a more productive agricultural activity (away from less productive home production) can lead to a significant reduction in poverty, even with little increase in wages. This holds especially when low-productivity home production is prevalent, as they observe in Bangladesh.[5] Overall, the case study findings confirm the positive effect of rising agricultural productivity (here instrumented through rainfall shocks) on poverty reduction. They further highlight the continuing importance of putting "surplus" labor in the form of underemployed and unemployed family labor, to more productive use and that this can be done both by reallocating it to nonagriculture (as discussed in the next point), but also to agriculture.

Seventh, reallocation of labor from agriculture, where most of the poor are currently located, to nonagriculture, which often is more skill intensive and where labor productivity tends to be higher, is an important channel through which poverty can be reduced. In many low-income countries (including in Africa) where food is primarily locally produced and consumed, increasing productivity growth in agriculture is often a precondition for releasing agricultural labor without creating hunger and starvation (Diao and McMillan). Yet, several structural factors further condition the speed and poverty reducing effects from such a reallocation of labor (including productivity growth outside agriculture) (Diao and McMillan). Eberhardt and Vollrath emphasize the importance of the elasticity of agricultural output with respect to labor in affecting the speed of structural transformation (and thus the potential contribution of labor reallocation to poverty reduction). In particular, they show that for an equivalent increase in agricultural total factor productivity (TFP), holding total agricultural output constant, economies with a low elasticity will be able to release a much larger amount of labor from agriculture to work in nonagriculture, than economies with a large elasticity. As a result, low elasticity economies will be able to develop faster than high-elasticities ones.[6] Employing panel data from 128 countries, they further show temperate and/or cold climate regions to have low, and tropical and highland regions to have much higher, agricultural output to labor elasticities. This provides an additional reason why structural change and development may have lagged in the latter. It further suggests that even larger increases in agricultural productivity will be needed to achieve the same effects as in temperate regions, or that progress will simply be slower.

Adam, Bevan and Gollin further emphasize that the benefits of public investment may at times be felt more in sectors other than those that are the primary target of the interventions. Differentiating the urban space in Tanzania between the capital, Dar, and secondary cities/towns, they find for example that increased public investment in urban areas in Tanzania, particularly in its secondary cities, would lead to larger welfare benefits for rural and unskilled (i.e. poor) households than agricultural biased investments of a similar size. When the productivity benefits of public investment occur in non-agricultural sectors, relative food prices fall by less and the incentives for out-migration are moderated. Put differently, it is the fall in food prices and the increase in labor supply in the urban locations that transfer the gains from agricultural-biased investments to urban households. They further show that because of the importance of the intersectoral labor and product

[4] Except for Malawi, the demand for agricultural wage labor remains limited in most African countries. Eighteen percent of rural households in African countries reports to have engaged in agricultural wage labor, contributing only 5 percent on average to total income, compared with 27 percent and 13 percent respectively in non-African countries respectively, and 29 percent and 16 percent respectively for Bangladesh (Davis, di Giuseppi, & Zezza, 2017).

[5] Divergence in the effects on agricultural wages and agricultural wage employment can for example arise if rising agricultural productivity pulls in labor from home production at a higher rate for the labor deficit than for the labor surplus households, which Emran and Shilpi show to be the case in Bangladesh.

[6] This abstracts from differences in their potential to affect population growth and growth in agricultural TFP, and assumes demand for agricultural products is income inelastic.

flows for the distributional effects of public investments and the intersectoral labor productivity differences, much poverty reduction is in effect to be expected from reducing transport costs, especially when targeted at eliminating rents in the sector, which accrue to the capitalist households otherwise.

Eight, a much ignored and under-appreciated point in the poverty reduction and structural transformation literature—and yet one central for real-world policy makers—is that the way public investments are financed[7] has first-order distributional implications which may overturn the underlying gains to productivity. An increase in public capital formation targeted to agriculture can, for example, negatively affect real consumption wages of the rural unskilled if financed from a tariff, but have a positive effect if financed through a consumption tax, which affects mainly the urban skilled (Adam, Bevan and Gollin). Great reliance on aid financed investment on the other hand, may cause real exchange appreciation, favoring the more traditional, domestic oriented nonagricultural sector over the more productive, open modern sector (Diao and McMillan). This would still come along with structural transformation and poverty reduction, as observed in much of Africa today, which saw a rapid expansion of its informal nonagricultural sector, because productivity in these non-tradable nonagricultural sectors is still greater than productivity in traditional agriculture. Given that more, and more unskilled, labor is hired by the informal sectors of this closed (nonagricultural) economy, it may also lead to larger poverty reduction than when public investment financing is less dependent on foreign grants, which would favor growth in the open modern economy and lead to greater labor productivity. The distributional consequences of aid based financing in the presence of a traditional, domestic in-between sector, or the sustainability of the growth model, are, however, not discussed as such in the paper in this issue.

3. Concluding remarks

The papers in this special issue confirmed that on average growth in agriculture tends to be more poverty reducing than an equivalent amount of growth outside agriculture. They further illuminated important nuances and complexities in these links between sectoral growth and poverty reduction, many of them mediated through the structural transformation. Finally, they drew attention to the differential distributional impacts of the financing modalities of public investments, which have remained much underappreciated so far. Yet, as we are starting to begin to understand the different links and effects, the ongoing digital revolution, the world's bifurcating demography and climate change continue to challenge our current metrics and methods, keeping the topic of agriculture, structural change and poverty reduction, a vibrant area of further investigation. We hope that the key insights emerging from the eight papers in this special issue will, at the very least, help us to keep our ignorance about the linkages and complexities in this area much better organized than in the past.

[7] The different financing mechanisms considered include deficit financing, which crowds out private investment, additional domestic taxation through indirect taxes on manufactured goods and services consumed by urban skilled households such as VAT and sales taxes or higher tariffs on manufactured imports as well as aid financing, which affects the real exchange rates and relative prices.

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